Monday, November 9, 2026

Boosting

It is an ensemble method where we aggregate the decision of several learners (classification / regression methods).

However, it differs from Bagging in several aspects.

Bagging	**Boosting**
• It is generally a variance reduction method	• Here we want to reduce the bias as well as the variance
• All observations always have equal weights	• Weights are adjusted at each step. Usually difficult observations get higher weight
• Several learners can run parallelly	• This is a sequential method, parallel processing cannot be done
• Final decision is obtained using majority voting	• Weighted voting is used for final decision. Classifiers (learners) with higher accuracy have higher weights

Discrete Adaboost (Freund & Schapire, 1998)

Consider a binary classification problem with the training data $\left\{(\underline{x}_i, y_i)\ ;\ i = 1, 2, \ldots, n\right\}$

where $y_i \in \{-1, 1\}$ for all $i = 1, 2, \ldots, n$

y_i: class label(Y_i)

$\underline{x}_i$: measurement vector / feature vector(X_i)

AdaBoost Algorithm

INITIALIZATION Adaboost algorithm starts with

$$F_0 = 0, \qquad W_0(i) = 1/n \quad \text{for } i = 1, 2, \ldots, n.$$

LOOP

For $m = 1$ to M (No. of classifiers to be aggregated)

Construct a classifier δ_m such that

$$\delta_m(x_i) \in \{-1, 1\} \quad \text{for all } i = 1, 2, \ldots, n.$$

Let

$$\varepsilon_m = \sum_{i=1}^{n} W_{m-1}(i)\, \mathbb{I}\{y_i \neq \delta_m(x_i)\} \Big/ \sum_{i=1}^{n} W_{m-1}(i)$$

be the misclassification rate of this classifier.

We define $\alpha_m = \frac{1}{2} \log\left(\frac{1-\varepsilon_m}{\varepsilon_m}\right)$ as the weight of the classifier δ_m.

and $F_m = F_{m-1} + \alpha_m \delta_m$

We readjust the weight as

$$W_m(i) = W_{m-1}(i) e^{-\alpha_m \delta_m(x_i) y_i} = \begin{cases} W_{m-1}(i) e^{-\alpha_m} & \text{if } x_i \text{ is correctly classified} \\ W_{m-1}(i) e^{\alpha_m} & \text{if } x_i \text{ is misclassified} \end{cases}$$

(We need $\varepsilon_m < 1/2$. So, any weak learner can be used as δ_m)

Note that $\varepsilon_m < 1/2 \Rightarrow \alpha_m > 0 \Rightarrow e^{-\alpha_m} < 1$ and $e^{\alpha_m} > 1$.

FINAL

DECISION Finally we classify an observation x depending on the sign of

$$F_M(x) = \sum_{m=1}^{M} \alpha_m \delta_m(x).$$

$$\delta_{\text{BOOSTING}}(x) = \text{sign}(F_M(x)) = \text{sign}\left(\sum_{m=1}^{M} \alpha_m \delta_m(x)\right)$$

- Why $\alpha_m = \frac{1}{2} \log\left(\frac{1-\varepsilon_m}{\varepsilon_m}\right)$?
- [At the m-th stage,] Boosting tries to minimize $E\left[e^{-Y F_M(X)}\right]$

$W_0(i) = 1/n$

$W_1(i) = \frac{1}{n} e^{-\alpha_1 \delta_1(x_i) Y_i}$

$W_2(i) = \frac{1}{n} e^{-\alpha_1 \delta_1(x_i) Y_i} \cdot e^{-\alpha_2 \delta_2(x_i) Y_i}$

$= \frac{1}{n} e^{-\sum_{j=1}^{2} \alpha_j \delta_j(x_i) Y_i}$

δ_1	$F_1 = \alpha_1 \delta_1$
δ_2	$F_2 = \alpha_1 \delta_1 + \alpha_2 \delta_2$
δ_3	$F_3 = \alpha_1 \delta_1 + \alpha_2 \delta_2 + \alpha_3 \delta_3$

$\delta_M(F_M) = \sum_{i=1}^{M} \alpha_i \delta_i$

At the m-th stage, Boosting tries to minimize $E\left[e^{-YF_M(X)}\right]$
An empirical version of this loss function is given by

$$\frac{1}{n}\sum_{i=1}^{n} e^{-Y_i F_m(X_i)}$$

α_m should be chosen such that this quantity is minimized.

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n} e^{-Y_i F_m(X_i)} &= \frac{1}{n}\sum_{i=1}^{n} e^{-Y_i\left(\sum_{j=1}^{m-1}\alpha_j\delta_j(X_i) + \underline{\underline{\alpha_m}}\delta_m(X_i)\right)} \\
&= \frac{1}{n}\sum_{i=1}^{n} e^{-Y_i\sum_{j=1}^{m-1}\alpha_j\delta_j(X_i)} \cdot e^{-Y_i\alpha_m\delta_m(X_i)} \\
&= \sum_{i=1}^{n} \underbrace{\left\{\frac{1}{n}e^{-Y_i\sum_{j=1}^{m-1}\alpha_j\delta_j(X_i)}\right\}}_{\text{What is it ?}} e^{-Y_i\alpha_m\delta_m(X_i)} \\
&= \sum_{i=1}^{n} W_{m-1}(X_i)e^{-Y_i\alpha_m\delta_m(X_i)}
\end{aligned}$$

$$F_M = \sum_{i=1}^{M}\alpha_i\delta_i$$

$$F_m(X_i) = \sum_{j=1}^{m}\alpha_j\delta_j(X_i)$$

$w(1)$	$w(2)$		$\cdots$	$w(n)$
1	2	3	$\cdots$	n
$1/n$	$1/n$	$1/n$	$\cdots$	$1/n$

Note that minimization of this is equivalent to minimization of

$$\frac{\sum W_{m-1}(X_i)e^{-Y_i\alpha_m\delta_m(X_i)}}{\sum W_{m-1}(X_i)} = \sum_{i=1}^{n}\widetilde{W}_{m-1}(X_i)e^{-Y_i\alpha_m\delta_m(X_i)}$$

normalized weights

$$\widetilde{W}_{m-1}(X_i) = W_{m-1}(X_i)\Big/\sum_{i=1}^{n} W_{m-1}(X_i)$$

so that $\sum_{i=1}^{n}\widetilde{W}_{m-1}(X_i) = 1.$

$$\begin{aligned}
&= \sum_{i=1}^{n}\widetilde{W}_{m-1}(X_i)e^{-Y_i\alpha_m\delta_m(X_i)} \\
&= \sum_{i:\, Y_i=\delta_m(X_i)}\widetilde{W}_{m-1}(X_i)\,e^{-\alpha_m} + \sum_{i:\, Y_i\neq\delta_m(X_i)}\widetilde{W}_{m-1}(X_i)\,e^{\alpha_m} \\
&= e^{-\alpha_m}(1-\varepsilon_m) \;+\; e^{\alpha_m}\varepsilon_m
\end{aligned}$$

α_m should be chosen by minimizing this quantity.

Differentiating w.r.t α_m and equating it with 0, we get

$$\begin{aligned}
& -e^{-\alpha_m}(1-\varepsilon_m) + e^{\alpha_m}\varepsilon_m = 0 \\
\Rightarrow\quad & e^{-\alpha_m}(1-\varepsilon_m) = e^{\alpha_m}\varepsilon_m \\
\Rightarrow\quad & \frac{1-\varepsilon_m}{\varepsilon_m} = e^{2\alpha_m} \qquad \Rightarrow \qquad \log\frac{1-\varepsilon_m}{\varepsilon_m} = 2\alpha_m
\end{aligned}$$

$$\Rightarrow \quad \alpha_m = \frac{1}{2}\log\frac{1-\varepsilon_m}{\varepsilon_m}$$

- Adaboost tries to minimize

$$\begin{aligned} E\left[e^{-YF_M(X)}\right] &= E\left\{E\left[e^{-YF_M(X)} \,\middle|\, \underline{X}\right]\right\} \\ &= E\left\{\underbrace{e^{-F_M(X)}p(1|X) + e^{F_M(X)}p(-1|X)}\right\} \end{aligned}$$

If we minimize

$$e^{-F_M(X)}p(1|X) + e^{F_M(X)}p(-1|X)$$

$$\begin{aligned} & e^{-\eta}p(1|x) + e^{\eta}p(-1|x) \\ & -e^{-\eta}p(1|x) + e^{\eta}p(-1|x) = 0 \\ \Rightarrow \quad & e^{-\eta}p(1|x) = e^{\eta}p(-1|x) \\ \Rightarrow \quad & \frac{p(1|x)}{p(-1|x)} = e^{2\eta} \qquad \Rightarrow \qquad \log\frac{p(1|x)}{1-p(1|x)} = 2\eta \end{aligned}$$

$$\log\frac{p(1|x)}{1-p(1|x)} = 2\underbrace{\left\{\sum_{m=1}^{M}\alpha_m\delta_m(x)\right\}}_{F_M(x)} \quad \longleftarrow$$

Boosting can be viewed as additive logistic reg. (Hastie & Tibshirani Ann Stat).

One can show that
this Adaboost algorithm can combine several weak learners to come up with a strong learner.

weak learner : a classifier with misclassification rate slightly smaller than 1/2 (slightly better than random guessing)

strong learner : given any $\varepsilon_0 > 0$, it can have misclassification rate smaller than ε_0 if we have enough observations

We have seen that

$$W_{m-1}(i) = \frac{1}{n}e^{-\sum_{j=1}^{m-1}\alpha_j\delta_j(X_i)Y_i} \quad \text{for all } i$$

$$\widetilde{W}_{m-1}(i) = \frac{W_{m-1}(i)}{Z_{m-1}} \qquad \text{where } Z_{m-1} = \sum_{i=1}^{n}W_{m-1}(i) = \sum_{i=1}^{n}\frac{1}{n}e^{-\sum_{j=1}^{m-1}\alpha_j\delta_j(X_i)Y_i}$$

- Result 1: The misclassification rate of the Adaboost classifier can be at most $\prod_{m=1}^{M} Z_m$

$$\left[\begin{aligned}\prod_{m=1}^{M} Z_m &= \prod_{m=1}^{M} 2\sqrt{\varepsilon_m(1-\varepsilon_m)} \qquad \longleftarrow \text{Result 2.}\\ &= \prod_{m=1}^{M} 2\sqrt{\left(\frac{1}{2}-\zeta_m\right)\left(\frac{1}{2}+\zeta_m\right)} \quad \text{where } 0<\zeta_m<1/2.\\ &= \prod_{m=1}^{M}\sqrt{1-4\zeta_m^2} \overset{\text{Check}}{\leq} \prod_{m=1}^{M} e^{-2\zeta_m^2}\end{aligned}\right]$$

Wednesday, November 11, 2026

Adaboost aggregates several weak learners (classifiers with misclassification rates smaller than $1/2$) to come up with a strong learner (that has misclassification rate smaller than ε for any given $\varepsilon > 0$).

Algo:

Starts with $W_0(i) = 1/n \quad \forall i = 1, 2, \ldots, n, \quad F_0(x) = 0 \quad \forall x \in \mathbb{R}^d, \quad y_i \in \{-1, 1\} \quad \forall i = 1, 2, \ldots, n$

For $m = 1$ to M

Run the classifier $\delta_m : \mathbb{R}^d \to \{-1, 1\}$

Compute the misclassification rate of δ_m

$$\varepsilon_m = \sum_{i=1}^{n} W_{m-1}(i)\, \mathbb{I}\{y_i \neq \delta_m(\underline{x}_i)\} \Big/ \sum_{i=1}^{n} W_{m-1}(i)$$

and $\alpha_m = \frac{1}{2}\log\left(\frac{1-\varepsilon_m}{\varepsilon_m}\right)$

Update the weights

$$W_m(i) = W_{m-1}(i)e^{-y_i\alpha_m\delta_m(x_i)} = \begin{cases} W_{m-1}(i)e^{-\alpha_m} & \text{if } \underline{x}_i \text{ is correctly classified by } \delta_m \\ W_{m-1}(i)e^{\alpha_m} & \text{if } \underline{x}_i \text{ is misclassified} \end{cases}$$

$F_m(x) = F_{m-1}(x) + \alpha_m\delta_m(x)$.

Finally, we classify an observation x to the class $\text{sign}(F_M(x))$

$$= \text{sign}\left(\sum_{m=1}^{M} \alpha_m\delta_m(x)\right)$$

[Note that $\alpha_m > 0$ whenever $\varepsilon_m < 1/2$, and $\alpha_m \uparrow$ as $\varepsilon_m \downarrow$]

Proof (weak learner $\longrightarrow$ strong learner)

Final classifier

$$\delta_{\text{BOOSTING}}(x) = \text{sign}(F_M(x)) = \text{sign}\left(\sum_{m=1}^{M} \alpha_m\delta_m(x)\right)$$

Misclassification rate (training sample misclassification rate)

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}_{\{y_i\neq \text{sign } F_M(x_i)\}} = \frac{1}{n}\sum_{i=1}^{n}\mathbb{I}_{\left\{y_i\neq \text{sign}\left(\sum_{m=1}^{M}\alpha_m\delta_m(x_i)\right)\right\}}$$
$$< \frac{1}{n}\sum_{i=1}^{n} e^{-\sum_{m=1}^{M}\alpha_m\delta_m(x_i)y_i} = \sum_{i=1}^{n}\left(\frac{1}{n}e^{-\sum_{m=1}^{M}\alpha_m\delta_m(x_i)y_i}\right)$$

$$W_0(i) = 1/n$$

$$W_1(i) = 1/n\, e^{-\alpha_1\delta_1(x_i)y_i} \longrightarrow \widetilde{W}_1(i) = \frac{1/n\, e^{-\alpha_1\delta_1(x_i)y_i}}{Z_1} \Rightarrow \frac{1}{n}e^{-\alpha_1\delta_1(x_i)y_i} = Z_1\widetilde{W}_1(i)$$

$$W_2(i) = \widetilde{W}_1(i)e^{-\alpha_2\delta_2(x_i)y_i} = \frac{\frac{1}{n}e^{-\alpha_1\delta_1(x_i)y_i}}{Z_1}e^{-\alpha_2\delta_2(x_i)y_i}$$

$$\widetilde{W}_2(i) = \frac{\frac{1}{n}e^{-[\alpha_1\delta_1(x_i)y_i+\alpha_2\delta_2(x_i)y_i]}}{Z_1Z_2}$$

$$\Rightarrow \quad \frac{1}{n}e^{-\{\alpha_1\delta_1(x_i)y_i+\alpha_2\delta_2(x_i)y_i\}} = Z_1Z_2\widetilde{W}_2(i)$$

$$\vdots$$

$$\Rightarrow \quad \frac{1}{n}e^{-\sum_{m=1}^{M}\alpha_m\delta_m(x_i)y_i} = \left(\prod_{m=1}^{M}Z_m\right)\widetilde{W}_M(i)$$

So, the training sample error rate of the Adaboost classifier has an upper bound given by

$$\sum_{i=1}^{n}\left(\prod_{m=1}^{M}Z_m\right)\widetilde{W}_M(i) = \left(\prod_{m=1}^{M}Z_m\right)\sum_{i=1}^{n}\widetilde{W}_M(i) = \left(\prod_{m=1}^{M}Z_m\right)$$

Note that

$$e^{2\alpha_m} = \frac{1-\varepsilon_m}{\varepsilon_m}$$

$$e^{\alpha_m}\varepsilon_m = (1-\varepsilon_m)e^{-\alpha_m}$$

So, the two quantities $\varepsilon_m e^{\alpha_m}$ and $(1-\varepsilon_m)e^{-\alpha_m}$ were equal.

$$\begin{aligned} Z_m &= \sum_{i=1}^{n} W_m(i) = \sum_{i=1}^{n} \widetilde{W}_{m-1}(i) e^{-\alpha_m \delta_m(x_i) y_i} \\ &= \sum_{i:\,\delta_m(x_i)=y_i} \widetilde{W}_{m-1}(i) e^{-\alpha_m \delta_m(x_i) y_i} + \sum_{i:\,\delta_m(x_i)\neq y_i} \widetilde{W}_{m-1}(x_i) e^{-\alpha_m \delta_m(x_i) y_i} \\ &= \left(\sum_{i:\,\delta_m(x_i)=y_i} \widetilde{W}_{m-1}(i) \right) e^{-\alpha_m} + \left(\sum_{i:\,\delta_m(x_i)\neq y_i} \widetilde{W}_{m-1}(x_i) \right) e^{\alpha_m} \\ &= (1-\varepsilon_m)e^{-\alpha_m} + \varepsilon_m e^{\alpha_m} \end{aligned}$$

Since these two terms are equal, we can write

$$Z_m = 2\sqrt{\varepsilon_m(1-\varepsilon_m)}$$

So, the training sample misclassification rate has an upper bound given by

$$\left(\prod_{m=1}^{M} Z_m \right) = \prod_{m=1}^{M} 2\sqrt{\varepsilon_m(1-\varepsilon_m)}$$

Since these learners are weak learners we have $\varepsilon_m = \frac{1}{2} - \gamma_m$ for some $\gamma_m > 0$ $(m = 1, 2, \ldots, M)$
So, the upper bound can be given by

$$\prod_{m=1}^{M} 2\sqrt{\left(\frac{1}{2}-\gamma_m\right)\left(\frac{1}{2}+\gamma_m\right)} = \prod_{m=1}^{M} \sqrt{1-4\gamma_m^2}$$

Note that $\sqrt{1-4\gamma_m^2} < e^{-2\gamma_m^2}$ (Check)

So, an upper bound can be given by $\prod_{m=1}^{M} e^{-2\gamma_m^2} = e^{-2\sum_{m=1}^{M}\gamma_m^2}$.

If $\sum_{m=1}^{\infty} \gamma_m^2 = \infty$, then this upper bound converges to 0 as M increases. and for $\varepsilon_0 > 0$, we can choose M_0 such that $e^{-2\sum_{m=1}^{M_0}\gamma_m^2} < \varepsilon_0$.

(This implies that the Adaboost classifier turns out to be a strong classifier)

[For instance if we can find an uniform lower bound γ^* for $\{\gamma_m : m \geq 1\}$

$$e^{-\gamma_m^2} < e^{-\gamma_*^2} \quad \forall m \quad \Rightarrow \quad e^{-2\sum_{m=1}^{M}\gamma_m^2} < e^{-M\gamma_*^2} \xrightarrow[\text{as } M\to\infty]{} 0$$]

α_m was obtained by minimizing

$$\underline{\varepsilon_m e^{\alpha}} + \underline{(1-\varepsilon_m)e^{-\alpha}} \quad \text{w.r.t } \alpha$$

which is equivalent to minimization of

$$\left(\sum_{i:\,\delta_m(x_i)\neq y_i} \widetilde{W}_{m-1}(i) \right) e^{\alpha} + \left(\sum_{i:\,\delta_m(x_i)=y_i} \widetilde{W}_{m-1}(i) \right) e^{-\alpha}$$

By minimizing, we get

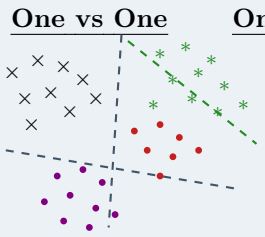
$$\alpha_m = \frac{1}{2}\log\left(\frac{1-\varepsilon_m}{\varepsilon_m}\right)$$

$$\frac{(1-\varepsilon_m)e^{-\alpha_m} + \varepsilon_m e^{\alpha_m}}{2} = \sqrt{(1-\varepsilon_m)\varepsilon_m}$$

$0 < \gamma_m < 1/2$
$0 < \gamma_m^2 < 1/4$
$0 < 4\gamma_m^2 < 1$

How to incorporate K classes ?

One option is pairwise classification followed by majority voting.



1 vs 2 2 vs 3
1 vs 3 2 vs 4
1 vs 4 3 vs 4

1 vs 2, 3, 4
2 vs 1, 3, 4
3 vs 1, 2, 4
4 vs 1, 2, 3

Adaboost algo for K classes (SAMME algo).

Start with $W_0(i) = \frac{1}{n} \quad \forall i = 1, 2, \ldots, n$

~~$F_0(x) = 0 \quad \forall x \in \mathbb{R}^d$~~ $\quad y_i \in \{1, 2, \ldots, K\}$ for all $i = 1, 2, \ldots, n$

For $m = 1$ to M

Run classifier $\delta_m : \mathbb{R}^d \to \{1, 2, \ldots, K\}$

Calculate

$$\varepsilon_m = \sum_{i=1}^{n} \mathbb{I}_{\{y_i \neq \delta_m(x_i)\}} W_{m-1}(i) \Big/ \sum_{i=1}^{n} W_{m-1}(i)$$

$$= \sum_{i=1}^{n} \widetilde{W}_{m-1}(i)\, \mathbb{I}_{\{y_i \neq \delta_m(x_i)\}}$$

$$\alpha_m = \frac{1}{2} \log \frac{1 - \varepsilon_m}{\varepsilon_m} + \log(K - 1)$$

Adjust the weights to get

$$W_m(i) = \begin{cases} W_{m-1}(i) e^{\alpha_m} & \text{if } x_i \text{ is misclassified by } \delta_m \\ W_{m-1}(i) e^{-\alpha_m} & \text{if } x_i \text{ is correctly classified by } \delta_m \end{cases}$$

Finally the classifier is given by

$$\delta_{\text{Boost}}(x) = \underset{1 \leq j \leq K}{\text{argmax}} \sum_{m=1}^{M} \alpha_m \mathbb{I}_{\{\delta_m(x) = j\}}$$